Support Operations Impact Analysis

Integration Gateway Adoption:

Workflow Continuity & Observability Requirements

Document Type	Support Operations Impact Analysis
Product	EchoCRS — Integration Gateway
Prepared By	Support Engineering
Audience	Product, Engineering, Support Leadership
Scenarios	5 current-state + future-state E2E analysis
Status	Draft — Pending Engineering Review
Version	1.0

Executive Summary

This document analyzes the operational impact of introducing the Integration Gateway as the unified connection layer between EchoCRS and property PMS systems. It is written from the support engineering perspective and examines five core support scenarios under the current architecture, documenting existing workflows, tool dependencies at each support tier, and the specific risks that emerge if the Integration Gateway does not maintain full observability into the message traffic it handles.

The central concern across all scenarios is log visibility. The current support model depends on end-to-end message tracing through Kibana and Splunk. If the Integration Gateway becomes an opaque connection point — passing traffic without surfacing that traffic to existing monitoring tools — Support loses the ability to triage, diagnose, and resolve issues at the L1 and L2 level. The result is a predictable increase in L3/Dev escalations, longer handle times, and degraded client experience during time-sensitive incidents.

A forward-state analysis and risk summary are included in the final section, along with architectural notes capturing open questions that require resolution before the Integration Gateway is deployed to production.

> ℹ **Notes**
>
> › This document reflects the current-state support architecture. Workflow steps and tool references describe how cases are handled today and what capabilities must be preserved — or equivalently replaced — in the Integration Gateway architecture.
>
> › Risk items marked with ⚠ represent operational gaps that will materialize if the stated capability is not maintained.

Section 1 — Current Architecture & Tool Inventory

Integration Flow (Current State)

All inbound and outbound traffic between the PMS and EchoCRS currently flows through the following path:

INBOUND R / PMS → Connector Throughway → EchoCRS → Distribution Channels (Web / GDS / OTAs)

OUTBOUND Distribution Channels (Web / GDS / OTAs) → EchoCRS → Connector throughway → R / PMS

Support Tool Inventory

The following tools underpin support workflows across all five scenarios documented in this analysis. Continuity of access to equivalent functionality is required in the Integration Gateway architecture.

Tool	Primary Use	Access Level
Kibana / Elastic	Primary tool for viewing inbound/outbound interface messages through the PMS Connector. Supports search by Echo Token, Property ID, Business ID, Restriction Code, Rate Code, Room/Inventory Code, Group Code, and Reservation ID.	**L1 / L2**
Splunk	Deep data querying for core messaging between PMS and EchoCRS. Used for root cause analysis and pending reservation review. Not available to L1 — restricted to L2/L3.	**L2 / L3**
Connectivity Test Tool	Used to check PMS endpoint reachability and transmit test data. Requires VPN access.	**L2 / L3**
Resend Tool	Reservation recovery tool specific to zero/one event scenarios. Provides monitoring of event queues with ability to resend entries in bulk.	**L2 / L3**
PMS Interface Report	In-product report used by Support and clients to determine whether reservations are transmitting successfully.	**L1 / L2 / Client**
PMS Interface Message Report	In-product report used by clients to view warnings and errors associated with ARI and reservation traffic.	**L1 / L2 / Client**
Reservations Report	Reservation list view noting modifications and cancellations. Includes cancellation-specific reporting.	**L1 / L2 / Client**

Section 2 — Current-State Use Case Scenarios

The following five scenarios represent the most common support cases handled through the PMS integration layer. Each scenario documents the current support workflow by tier and identifies the operational risks introduced if Integration Gateway traffic is not surfaced to existing monitoring tools.

Scenario 1 — Reservation Overbooking

A property reports that a reservation was accepted despite no available inventory. Support must trace inventory messages relative to the booking timestamp to determine whether the overbooking is valid, and identify root cause if so.

Current Support Workflow

Role	Actions
L1	1. Locate the reported overbooking record in Kibana with timestamp. 2. Locate the last inventory message in Kibana received prior to the overbooking, with timestamp, for comparison. 3. For additional insight, locate the first inventory message after the overbooking to determine if the next message is 0 or negative. 4. Check for cancellations and modifications using the Reservations Report.
L2	1. Consult with L1 to confirm findings. 2. Confirm property configuration settings. 3. Review Splunk to locate pending reservations if applicable. 4. Determine whether case can be resolved and returned to client, or requires L3 escalation.
L3	If escalated, conduct further review via Splunk to identify root cause and resolution.

⚠ Risk Without Full Integration Gateway Visibility

- Without end-to-end logs through the Integration Gateway, if an inventory message is not found in Kibana, Support cannot validate whether the message was received from the PMS relative to the booking timestamp.
- All potential overbooking cases where messages cannot be found will require L3/Dev escalation to answer: Was the inventory message received and at what time? Why was the message not passed through to the connector layer?
- Result: Increased L3/Dev handle time and delayed resolution for time-sensitive reservation disputes.

Scenario 2 — ARI Discrepancy (PMS-Centric)

A property submits Availability, Rate, or Inventory (ARI) updates from the PMS, but the updates are not reflected in EchoCRS. This scenario covers the case where the update originates from the PMS side and may involve a mapping error, a rejected message, or a connection failure.

Client Self-Service Path

Role	Actions
Client	1. Reviews the PMS Interface Messages Report in EchoCRS to view warnings and errors preventing the update. 2. Uses report information to determine next steps: correct configuration in EchoCRS or PMS, contact EchoCRS Support, or contact PMS Vendor.

⚠ Risk Without Full Integration Gateway Visibility

- If the Integration Gateway introduces intermittent or sustained connection failures, in-product reporting may be interrupted. Clients lose self-service capability, leading to increased inbound case volume.

Support Escalation Path

Role	Actions
L1	1. Locate the most recent ARI message in Kibana with timestamp. 2. Review for warnings or errors returned (e.g., codes do not match, rates are linked, processing error, etc.). 3. Advise client on next steps based on findings.
L2	1. Consult with L1 to confirm findings. 2. Confirm settings or advise on warnings/errors not recognized by L1. 3. If these are not the root cause, conduct further review and confirm L3 escalation if needed.
L3	If escalated, conduct further review via Splunk to identify root cause and resolution.

⚠ Risk Without Full Integration Gateway Visibility

- L1 and clients use in-product screens to resend and retry reservations. Without end-to-end logs, Support cannot validate whether messages transmitted through all connection points (EchoCRS → Connector → Gateway → PMS). Support visibility currently ends at Kibana.
- If in-product screens are blank and messages are not found in Kibana, Support cannot validate message receipt. All such cases require L3/Dev escalation to answer: Were messages received from the PMS and at what time? Why was the message not passed through the connector layer?
- Result: Increased escalation volume and handle time across both self-service and support-assisted paths.

Scenario 3 — ARI Not Received (CRS-Centric)

A property's ARI updates are not reaching PMS from the CRS side. This scenario covers cases where the failure originates within the CRS or channel configuration rather than the PMS.

Support Escalation Path

Role	Actions
L1	1. Check configuration in EchoCRS and the Interface Configuration to confirm correct setup. 2. Locate the most recent ARI message in Kibana with timestamp. Review for warnings or errors. 3. Advise client on next steps based on findings (e.g., code mismatch, processing error, etc.).
L2	1. Consult with L1 to confirm findings.

	2. Confirm settings or advise on unrecognized warnings/errors. 3. If unresolved, conduct further review. Engagement of PMS vendor may be necessary.
L3	Based on vendor findings, conduct further review via Splunk to identify root cause and resolution.

⚠ Risk Without Full Integration Gateway Visibility

- Without end-to-end logs through the Integration Gateway, Support cannot validate message flow across all connection points (EchoCRS → Connector → Gateway → PMS). Visibility ends at Kibana.
- Cases where the failure point cannot be determined from Kibana alone will require L3/Dev escalation to validate transmission end-to-end.
- Result: Increased L3/Dev case volume and handle time, including during time-sensitive incidents.

Scenario 4 — Zero / One Event (Reservation Transmission Failure)

Reservations are not reaching the PMS (zero events — no transmission attempt) or are transmitting but not receiving a response from the PMS (one event — sent but unacknowledged). These cases range from property-specific issues to system-wide queue failures.

Client Self-Service Path

Role	Actions
Client	1. Notices reservations are not reaching the PMS during routine review. 2. Reviews the PMS Reservations Report and attempts to resend. 3. If no events are generated or no PMS response received, reviews the PMS Interface Messages Report for warnings/errors. 4. Uses report information to retry, correct configuration, contact PMS vendor, or contact EchoCRS Support.

⚠ Risk Without Full Integration Gateway Visibility

- In-product reporting requires a maintained connection to function. Any connection failure introduced by the Integration Gateway will disable client self-service for zero/one event detection, or can result in one events being missed by support teams as transmissions will appear to be delivered through the connector layer and the integration gateway layer is the failure point — whether for a single reservation or a property-wide outage. Result: Increased inbound case volume.

Support Monitoring & Recovery Path

Role	Actions
L1	1. Review PMS Interface Report and attempt to resend when zero/one events are reported.

	2. For zero events: if reservations are not sent, check Kibana for errors. If no errors found, engage L2. 3. For one events: if reservations are sent but no PMS response is received, check Kibana for errors, inform client, and advise to engage PMS vendor.
L2	1. For zero events: access the Resend Tool and search for zero events for the current date. 1,000+ entries indicates a system-wide issue. 2. Attempt recovery by resending reservations — processing can take up to 30 minutes. 3. Engage NOC if entries do not decrease or continue to climb.
L2 / NOC	1. NOC provides additional monitoring of backend queues. 2. A bridge may be opened for faster engagement and resolution of traffic processing.

⚠ Risk Without Full Integration Gateway Visibility

- Without a tool equivalent to the Resend Tool for the Integration Gateway, zero event cases cannot be monitored or recovered at L2. Cases must escalate through L1 → L2 → L3/NOC to determine whether the issue is property-specific or system-wide — significantly increasing resolution time and interface downtime for the property.
- For one events: without end-to-end logs, Support cannot validate that reservations passed through all connection points. L3/Dev must validate transmission before Support can advise the client to engage their PMS vendor — adding handle time during time-sensitive incidents.

Scenario 5 — High ARI Volume / System Degradation

Unusually high ARI message volume from one or more properties is contributing to system degradation. NOC detects the issue and engages L2 to identify and mitigate the source.

Support & NOC Response Path

Role	Actions
L2 / NOC	NOC engages L2 regarding elevated ARI volume detected in system monitoring.
L2	1. Reviews Splunk dashboard to identify and confirm potential sources ('bad actors') contributing to system degradation. 2. Identifies traffic spikes using Kibana charts and logs, along with any warnings or errors in the messages sent. 3. Engages affected clients to inquire about recent configuration changes that may explain the volume spike. 4. Requests configuration adjustments to optimize traffic flow and reduce processing load. 5. Repeatedly reviews Kibana charts and logs to confirm adjustments are taking effect. 6. Reviews Splunk dashboard to confirm the client is no longer contributing to degradation.

⚠ Risk Without Full Integration Gateway Visibility

- Without a monitoring tool equivalent to Kibana for the Integration Gateway layer, L2 cannot independently identify traffic sources or spikes. All identification and mitigation steps must be routed through L3/NOC.
- Each configuration change must run through higher support tiers for validation, compounding delays during an active system degradation event.
- Result: Extended system degradation windows and increased time-to-mitigation.

Section 3 — Future-State Analysis: Integration Gateway E2E Flow

This section addresses the architectural implications of introducing the Integration Gateway as the unified connection point between EchoCRS and all PMS systems. It documents what must be true about the Integration Gateway's observability and tooling integration to preserve current-state support capabilities.

Proposed Future-State Flow

Phase 1 — Gateway as Added Connection Point

EchoCRS → PMS Connector → Integration Gateway → PMS

All existing monitoring tools must receive traffic feed from all connection points, including the Gateway layer.

If the Integration Gateway is introduced as an additional connection point on the throughway, the following capability is required to maintain current support operations:

- All message traffic passing through the Integration Gateway must be pushed to Kibana and Splunk for visibility at L1/L2/L3 respectively.
- The Gateway must surface timeout events, retry loops, and hung traffic — with bidirectional recovery capability (ability to push traffic in either direction from the Gateway).
- If the Gateway becomes a point of failure, Support must be able to identify this from existing tools without requiring L3/Dev intervention for basic triage.

Consolidated Risk Summary

The following risks apply if the Integration Gateway does not maintain full observability into message traffic:

#	Risk	Impact
1	Issues arising at the Integration Gateway layer will produce gaps in troubleshooting and root cause analysis. Support cannot see errors, failures, or timestamps at the Gateway.	*Misdiagnosis; L3 escalation required for basic triage*
2	Connection failures at the Gateway cannot be validated — timeout, system failure, authentication errors are all invisible without Gateway-level logs.	*Inability to confirm failure origin; incorrect system blamed*
3	Latency within the Gateway cannot be measured, which may result in an incorrect system being identified as the performance bottleneck.	*Inefficient optimization; client misdirected*

4	The PMS Connector may not reflect accurate error messages when a failure originates in the Gateway layer, leading Support to follow incorrect troubleshooting paths.	*Incorrect resolution steps; repeat contacts*
5	All of the above contribute to increased case volume, increased escalation rates, and longer handle time — including during time-sensitive incidents.	*Degraded SLA performance; client experience impact*

Open Architectural Questions

The following questions require resolution from Product and Engineering before the Integration Gateway architecture can be finalized for support operations:

#	Open Question
1	Splunk architecture: One instance is planned between the Gateway and PMS. Potentially 5 instances (one per Product) between the Product and Gateway, replacing Elastic/Kibana logging. Confirm final instance count and ownership.
2	Operational requirement: Kibana should be placed as a layer on top of Splunk to maintain a user-friendly troubleshooting interface for L1 Support. Confirm whether this will be implemented.
3	Gateway team scope: The Integration Gateway team will address questions about how the Gateway interacts with the PMS. Product-side questions remain the responsibility of individual Product teams. Confirm escalation path boundaries.
4	Single Screen availability: It is not yet confirmed whether Single Screen tooling will be used for the Integration Gateway, or whether the current model will apply.
5	Customer partitioning model: How will properties be partitioned within the Integration Gateway? By region, customer size, or another model? This affects how Support searches and filters Gateway logs.
6	OHIP / Oracle mapping: If OHIP properties are onboarded, mapping tables will be required since Oracle will no longer perform mappings directly. Confirm Support's role in this workflow.

This document is confidential and intended for internal product, engineering, and support leadership review only. It reflects the support engineering perspective on Integration Gateway requirements and does not represent a final architectural decision. All open questions should be resolved in coordination with Product and Engineering before Gateway deployment.